



NeuroSpace

Complete evaluation export

Job ID	c97883bb-61af-4977-9e98-6a28d1e787b9
Status	completed
Result timestamp	2026-06-07T00:46:03.771879
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Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	6
class_names	1, 2, 3, 4, 5, 6
qrnn_backend	qiskit
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1

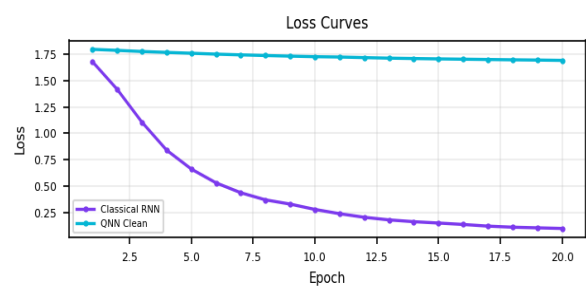
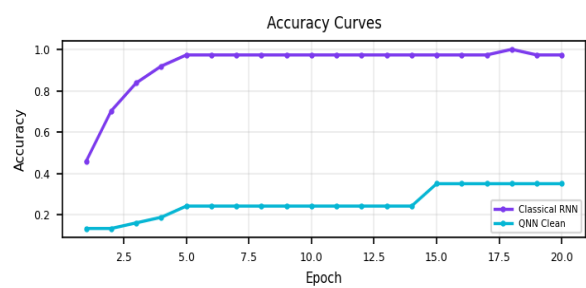
Best Model

Winner	RNN Accuracy	QNN Accuracy	Improvement
RNN	94.59%	31.08%	63.51%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss	Training Time
Classical RNN	94.59%	95.44%	94.59%	94.28%	0.0970	6.28s
QNN Clean	31.08%	39.49%	31.08%	29.97%	1.6926	21.73s

Learning Curves



Classical RNN

Metric	Value
Accuracy	94.59%
Precision	95.44%
Recall	94.59%
F1 Score	94.28%
Loss	0.0970
Training Time	6.28s
Train Loss	0.1416
Val Loss	0.0970
Train Accuracy	97.25%
Val Accuracy	97.30%
Train Size	255
Val Size	37
Test Size	74

Classical RNN Confusion Matrix

Actual / Predicted	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Class 1	23	0	0	0	0	0
Class 2	0	8	0	2	1	1
Class 3	0	0	15	0	0	0
Class 4	0	0	0	10	0	0
Class 5	0	0	0	0	10	0
Class 6	0	0	0	0	0	4

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
1	100.00%	100.00%	100.00%	23.0000
2	100.00%	66.67%	80.00%	12.0000
3	100.00%	100.00%	100.00%	15.0000
4	83.33%	100.00%	90.91%	10.0000
5	90.91%	100.00%	95.24%	10.0000
6	80.00%	100.00%	88.89%	4.0000
macro avg	92.37%	94.44%	92.51%	74.0000
weighted avg	95.44%	94.59%	94.28%	74.0000
Accuracy			94.59%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.7791	1.6800	33.33%	45.95%
2	1.3591	1.4181	56.08%	70.27%
3	1.0211	1.1066	73.73%	83.78%
4	0.8008	0.8407	85.88%	91.89%
5	0.6374	0.6610	90.59%	97.30%
6	0.5612	0.5300	89.41%	97.30%
7	0.4649	0.4363	90.98%	97.30%
8	0.4427	0.3686	90.98%	97.30%
9	0.3563	0.3285	94.12%	97.30%
10	0.3398	0.2778	94.90%	97.30%
11	0.3063	0.2372	93.73%	97.30%
12	0.2653	0.2032	95.69%	97.30%
13	0.2420	0.1784	94.51%	97.30%
14	0.2204	0.1614	97.25%	97.30%
15	0.1818	0.1487	97.25%	97.30%
16	0.2003	0.1350	96.08%	97.30%
17	0.1776	0.1194	96.47%	97.30%
18	0.1361	0.1091	98.04%	100.00%
19	0.1357	0.1031	98.43%	97.30%
20	0.1416	0.0970	97.25%	97.30%

QNN Clean

Metric	Value
Accuracy	31.08%
Precision	39.49%
Recall	31.08%
F1 Score	29.97%
Loss	1.6926
Training Time	21.73s
Train Loss	1.6221
Val Loss	1.6926
Train Accuracy	36.08%
Val Accuracy	35.14%
Train Size	255
Val Size	37
Test Size	74

QNN Clean Confusion Matrix

Actual / Predicted	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Class 1	5	2	14	0	2	0
Class 2	0	3	0	5	4	0
Class 3	2	12	0	1	0	0
Class 4	0	2	0	6	1	1
Class 5	0	0	0	2	8	0
Class 6	0	1	0	0	2	1

QNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
1	71.43%	21.74%	33.33%	23.0000
2	15.00%	25.00%	18.75%	12.0000
3	0.00%	0.00%	0.00%	15.0000
4	42.86%	60.00%	50.00%	10.0000
5	47.06%	80.00%	59.26%	10.0000
6	50.00%	25.00%	33.33%	4.0000
macro avg	37.72%	35.29%	32.45%	74.0000
weighted avg	39.49%	31.08%	29.97%	74.0000
Accuracy			31.08%	

QNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8151	1.7975	18.04%	13.51%
2	1.7951	1.7877	18.04%	13.51%
3	1.7735	1.7771	17.25%	16.22%
4	1.7604	1.7685	22.75%	18.92%
5	1.7447	1.7605	28.63%	24.32%
6	1.7323	1.7520	30.98%	24.32%
7	1.7242	1.7450	30.20%	24.32%
8	1.7082	1.7388	29.80%	24.32%
9	1.7002	1.7330	37.25%	24.32%
10	1.6863	1.7279	37.25%	24.32%
11	1.6776	1.7246	36.47%	24.32%
12	1.6721	1.7190	36.08%	24.32%
13	1.6617	1.7138	35.69%	24.32%
14	1.6554	1.7103	36.08%	24.32%
15	1.6491	1.7068	36.47%	35.14%
16	1.6488	1.7038	36.86%	35.14%
17	1.6354	1.7010	36.86%	35.14%
18	1.6337	1.6981	36.08%	35.14%
19	1.6301	1.6958	36.08%	35.14%
20	1.6221	1.6926	36.08%	35.14%